
EY
Building a better working world
2024
Campus
Finalists

ReviveSpark Tech : *Turning Waste into Resource*

The IDEA

Rapidly exploding EV and electronics industry → Need for energy specifically batteries → Reduce reliance on imports and rampant production expansion → **Lithium Ion Battery Recycling**

WHY : Issues & Importance

100 million
vehicle **batteries** are expected to be **retired** in the next decade; waste management challenges

Waste Management
Only 5% of lithium-ion batteries are recycled; the rest contribute to landfill waste.

Resource Depletion
Lithium demand may reach 1.5 million tons by 2025, risking material shortages.

Economic Concerns
High production costs and reliance on mined materials expose the EV market to volatility.

Toxic Contamination
Hazardous materials from batteries threaten soil and water.

Supply Chain Risks
Heavy reliance on imports, especially for cobalt, raises supply chain vulnerabilities.



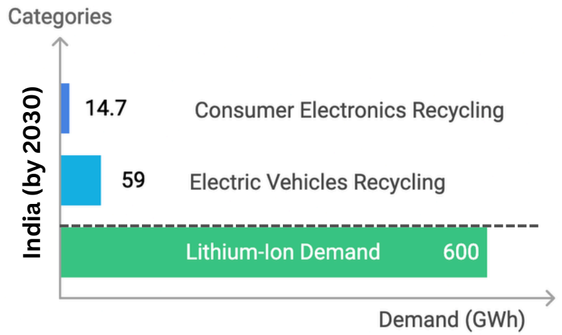
Market Landscape & Opportunity

$1.5 billion — **2023**: Li-ion (LIB) market size India

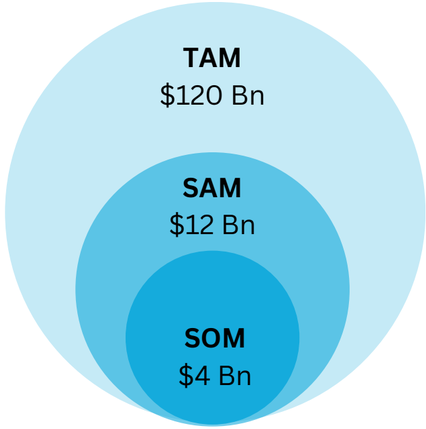
$12 billion — **2040**: Li-ion (LIB) market size India

35% CAGR

$150 billion — **2040:** Li-ion (LIB) market size global



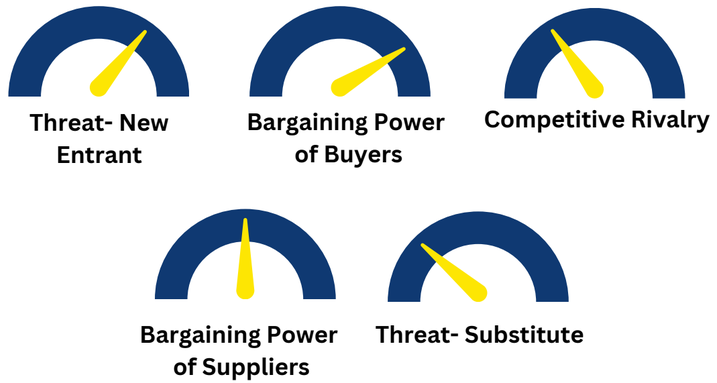
Market Share (By 2040)



Target Group

- **Energy Companies**
- **Automobile & Battery firms**
- **B2C for Individuals**
- **Electronics Companies**

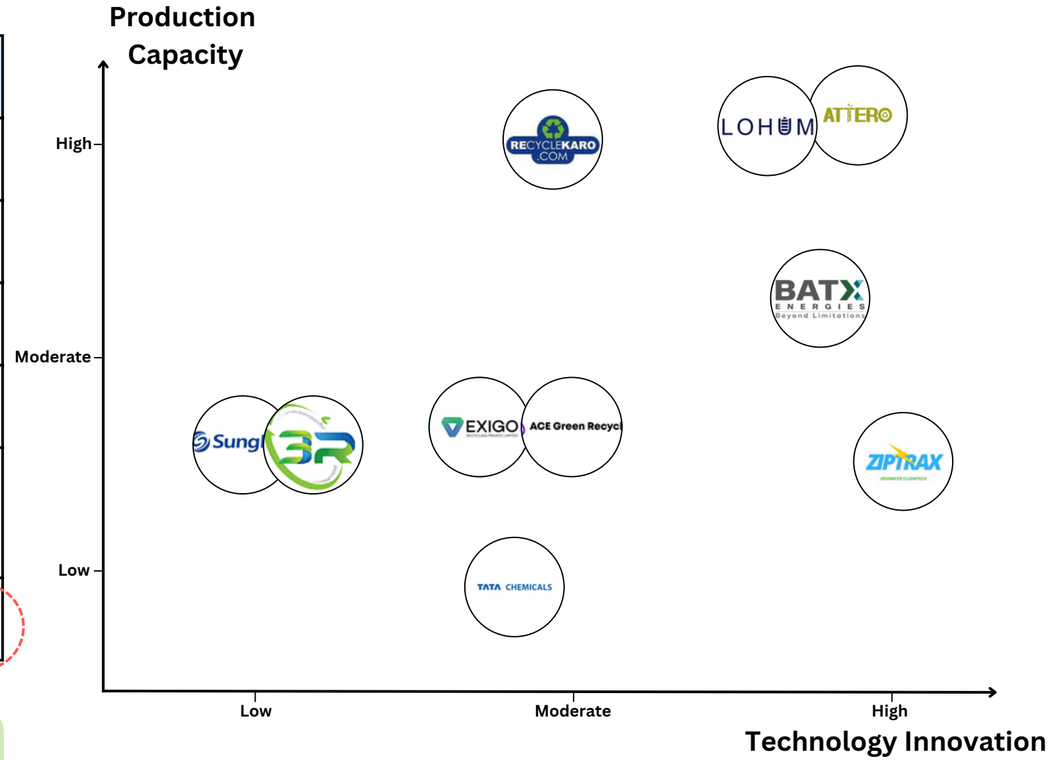
Market Entry Analysis



Growth Drivers



Company	Attero	Lohum	BatX Energies	Exigo Recycling	Tata Chemicals	RecycleKaro
Annual Capacity	15,000 tons	10,000 tons	15,000 tons	10,000 Tons	500 tons	2,500 tons
Key Partnerships	MG Motors, Hyundai, Tata	Glencore, Mercedes-Benz	Local sourcing networks	IT and electronics companies	Tata Motors, ISRO	Bajaj Auto, HeroMotoCorp
Technology Used	Hydrometallurgical	NEETM technology	Hydrometallurgical	Various	Chemical recovery	Hydrometallurgical
Location	Roorkee, Uttarakhand	Noida	Sikandrabad, Uttar Pradesh	NCR	Mumbai	Palghar, Maharashtra
Core Strengths	High recovery rates, 47 patent portfolio	Integrated repurpose and recycler, NEETM tech patent	Zero emissions, innovative recycling process	Broad e-waste focus, PRO compliance	High purity recovery, strong brand	High efficiency and purity, large scale plans
Weaknesses	High R&D costs	Limited global presence	Newer company, smaller scale	More focused on e-waste	Scaling from pilot phase	High initial R&D investment



Market GAPS

- Reverse Logistics Network
- Integration with Renewable Energy Projects
- End-to-End Service Offerings
- Transparency and Traceability

Learnings

- Integration with OEMs is critical for securing steady input materials.
- Investing in technology & patents = competitive edge
- Both recycling and refurbishing = competitive edge
- Scalability is crucial; core ops and reverse logistics with partnerships.

Obstacles

- High upfront costs for R&D and technology.
- Market concentration & geo-limitations.
- Scaling challenges from pilot to commercial production.
- Strict regulatory compliance (environmental, international, and local).

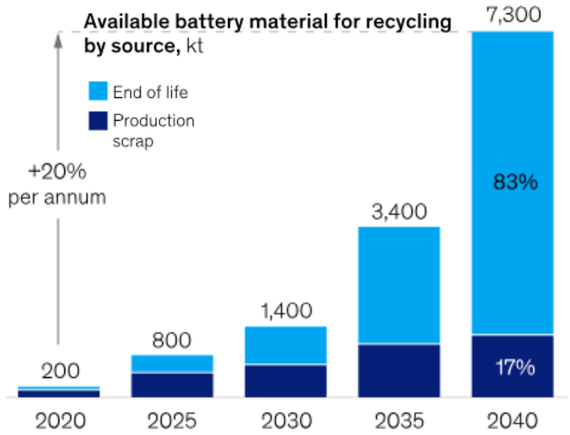
Key Recommendations:

1. **Specialisation**:
   - Specialize in specific battery types or regional markets.
   - *Integrated* recycling and refurbishing.
2. **Strategic Partnerships**:
   - Collaborate with *established players* for used battery
   - Leverage existing *technologies, supply chains*, and *markets*.
3. **Innovative Business Models**:
   - Offer value-added services like battery life extension.
   - Explore opportunities in energy storage solutions.

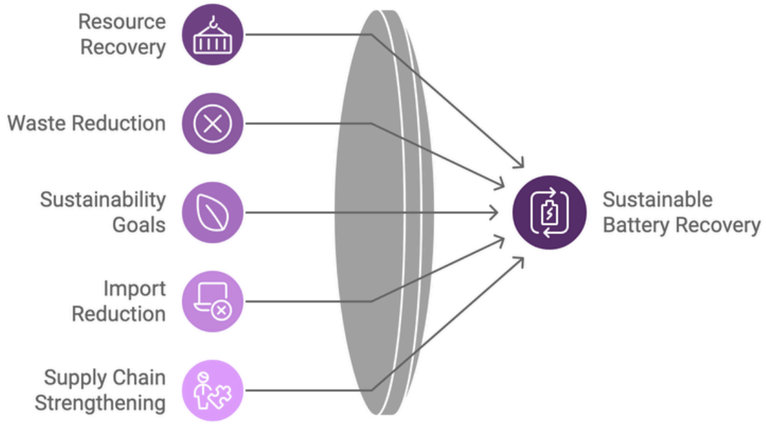
Rationale

- **Market Entry Opportunities** — Feasible entry for new players
- **Government Policies** — Supportive regulations and initiatives
- **Consumer Electronics** — Rising demand for portable power sources
- **Battery Energy Storage Systems** — Increasing need for energy storage solutions in renewable
- **Electric Vehicle Sector** — Growth in EVs driving battery demand and charging stations

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People

90% **less landfill waste due to recycling**

60% **energy savings**; recycling vs new battery production

Best case scenario by 2040.

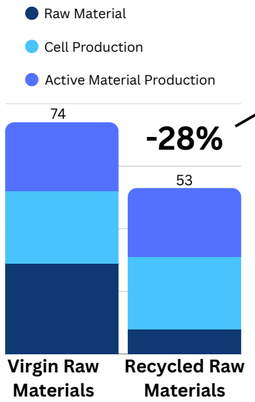
Annual water use avoided (in acre feet)
224,000

Land Use Avoided (in acres)
35,937

Number of Jobs created in the United States
67,029

Assuming up to 2.5 full-time employees per kiloton of recycling capacity. Water and land intensity of mining depends on location and critical mineral in question.

Planet



lower carbon footprint /kWh of battery cell with recycled material than newly mined ones.

50% **lower emissions** in battery production

Profit

95% **recovery** of **valuables** (lithium, cobalt, & nickel) from used batteries

35% reduced **dependence** for **lithium** & **cobalt**, boosting local supply

Government Goals

National Action Plan on Climate Change (NAPCC)

- National Solar Mission
- National Mission for Enhanced Energy Efficiency
- National Mission on Sustainable Habitat

Reduced Imports and Supply Chain Strengthening

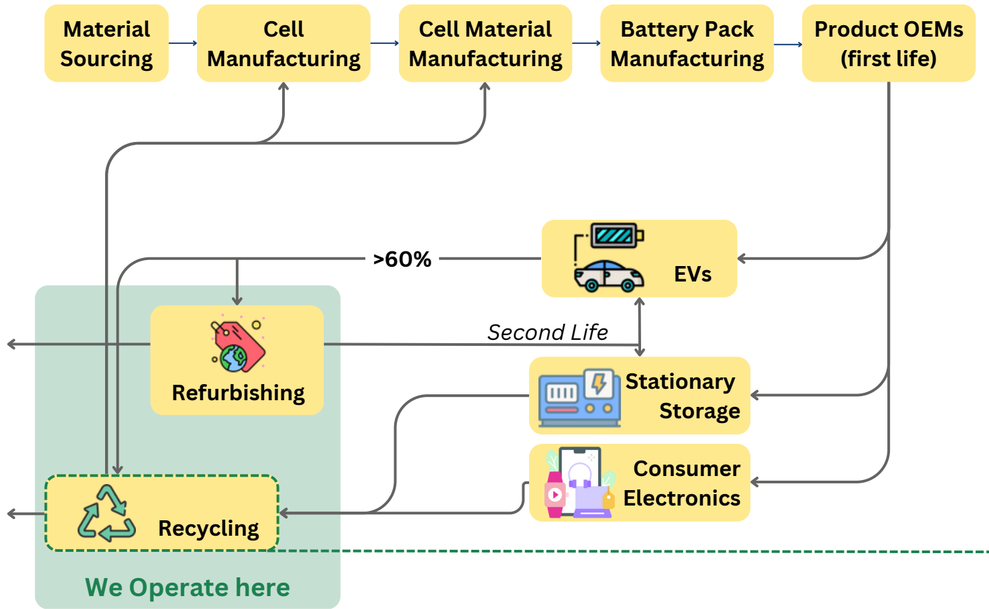
Government Initiatives Supported

- Extended Producer Responsibility (EPR)
- Make in India
- Production Linked Incentive (PLI) Scheme
- Atmanirbhar Bharat
- Resource Efficiency and Circular Economy Industry Coalition (RECEIC)

Mckinsey | MeitY

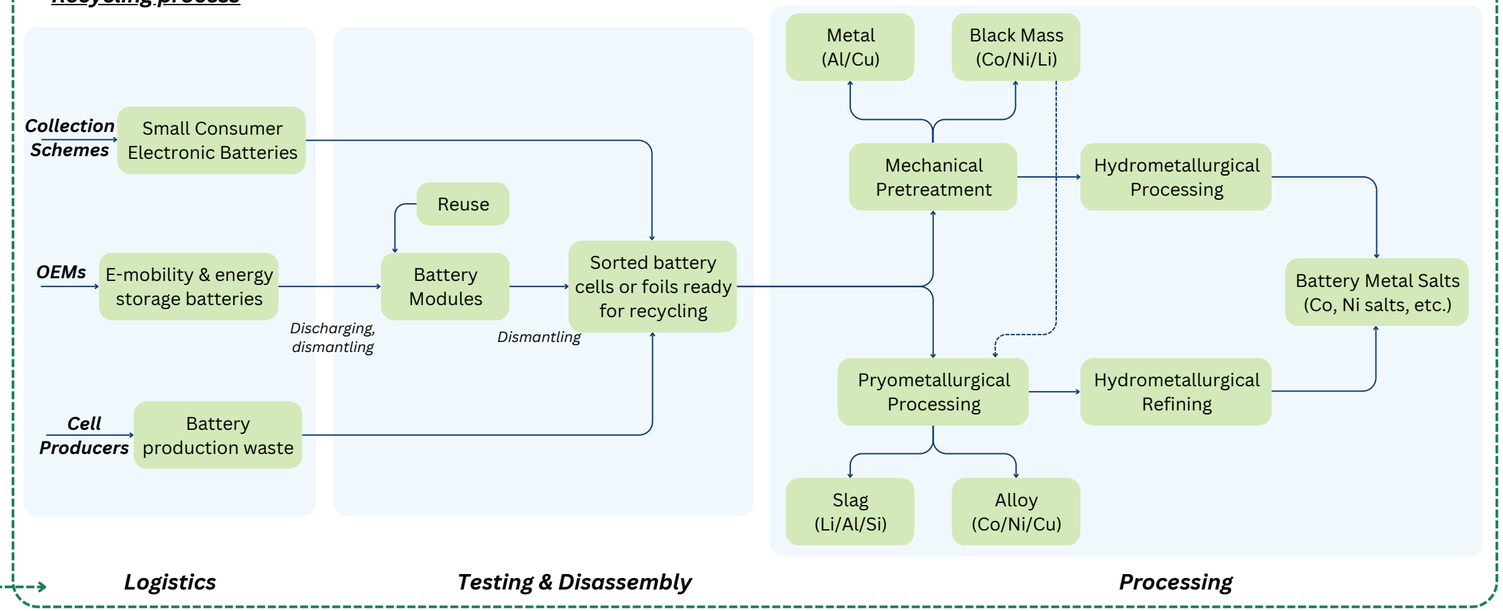
Value Chain of Battery Manufacturing

(Niti Ayog)



Recycling process

(McKinsey)



Recycling Process Step	Description	Relevant Government Schemes/Initiatives
Collection Schemes	Collection from consumers and OEMs.	E-Waste Management Rules, 2016; Extended Producer Responsibility (EPR); Swachh Bharat Mission
Testing & Disassembly	Checking battery integrity and dismantling.	FAME India Scheme; Mission LiFe; RECEIC
Mechanical Pretreatment	Separating metals and preparing black mass.	National Electric Mobility Mission; RECEIC; Mission LiFe
Pyrometallurgical Processing	Using high temperatures to extract metals.	Make in India (for metallurgical industries)
Hydrometallurgical Processing	Chemical solutions to leach metals from black mass.	Atal Innovation Mission; India's National Policy on Resource Efficiency
Hydrometallurgical Refining	Purifying recovered metals to battery-grade materials.	Digital India; Science, Technology and Innovation Policy
Reuse	Repurposing batteries for secondary applications.	Integrated Power Development Scheme; Smart Cities Mission; Mission LiFe

Impacts

- Investment Incentives
- Govt. Support
- Circular Economy Adoption
- Competitive Adv.

Innovations in Battery Recycling:

- **Direct Recycling:** Recovering cathode and anode; preserving structure and composition.
- **Bioleaching:** Eco-friendly metal extraction using *microorganisms*.

AI in Battery Recycling:

- **Intelligent Sorting:** Accurate identification and sorting.
- **Predictive Maintenance:** Optimized equipment lifespan.
- **Process Optimization:** Maximized metal recovery and efficiency.

Business Model	Market Demand	Scalability	Capital Intensity	Regulatory Support	Innovation Potential	Overall Viability
Refurbishing	7	8	6	8	7	522
Recycling	9	8	7	9	7	648
Manufacturing (Assembly)	8	5	8	6	4	120
Manufacturing (Proper)	8	3	10	6	3	43.2
Battery as a Service (BaaS)	6	9	8	5	8	270
Carbon Credits Trade	8	7	6	8	6	448
Consultation Services	9	9	5	7	6	680

$$\text{Overall Viability} = \frac{\text{Demand x Scalability x Regulatory Support x Innovation}}{\text{Capital Intensive}}$$

Brand Mascot

Voltie



- **Educational Content**: Develop videos featuring Voltie explaining recycling and sustainability.
- **Marketing & Events**: Utilise Voltie in marketing materials and at events.
- **Community Outreach**: Engage schools and communities through Voltie-led educational programs.



GTM Strategy

Competitive Advantage

Cost | **Geographic** | **Focus** | **Innovation**

Differentiation | **Speed** | **Customer Service**

Consulting Services
- Offer expertise in battery waste management and regulatory compliance.

Clients: Companies looking to improve sustainability practices.

Carbon Credits Trading
- Earn and trade credits for helping others meet EPR obligations.

Clients: Fossil fuel based, manufacturing, services and transportation.

Streamlined Logistics:
- Quick pickup and return of batteries (partners)
- Software to optimise routes and **reduce TAT**.

Rapid Processing Technology:
- Explore options and invest in patents

Tailored Solutions:
- Flexible pricing models
- Personalized plans
- Dedicated support

Transparency:
- Offer clear reporting on environment impact, savings

Subscription Model
- Regular collection and recycling services on a subscription basis.

Clients: Data centers, telecom companies, and transport operators.

Material Sales | **Service Fees** | **Refurbished Product Sales**

Revenue Models

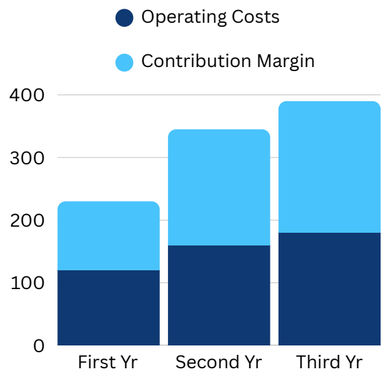
Total waste India: 98,875 tons/yr

Capacity 10,000 tonnes/yr

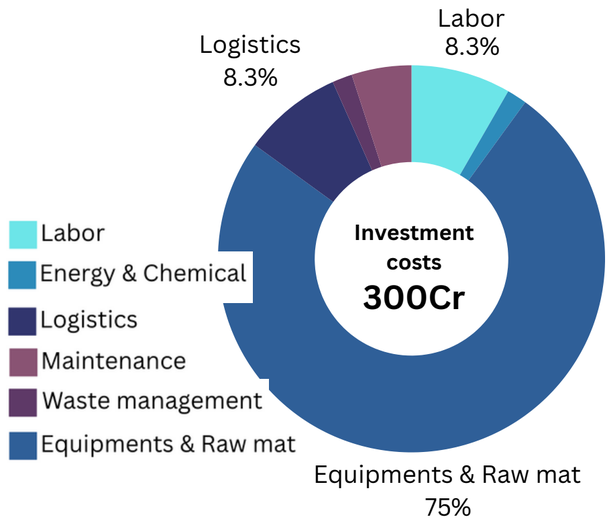
CAGR 35%+ till 2035

Profitable 4th yr onwards

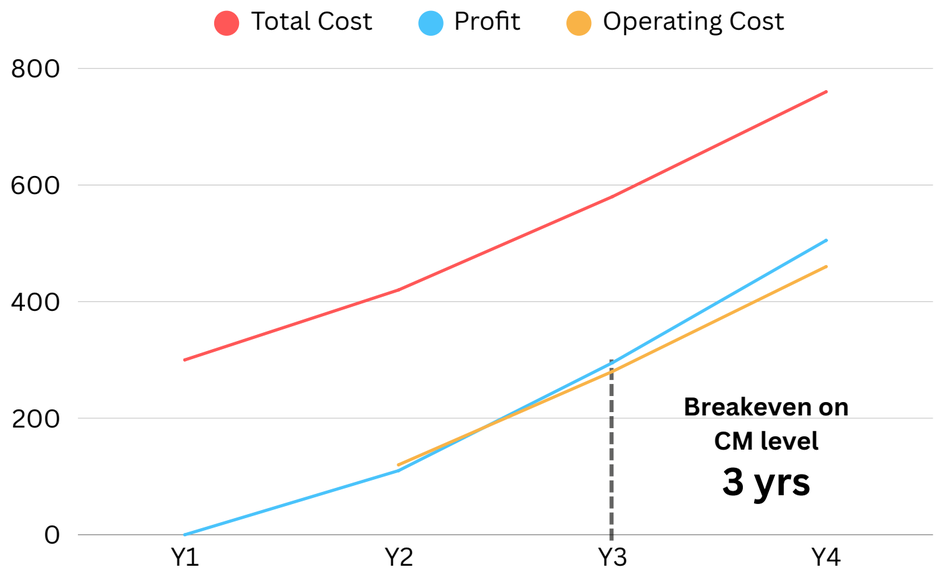
Year (Capacity)	Revenue from Recovered Material (Rs/tonne)	Variable Costs (Rs/tonne)	Contribution margin (Rs/tonne)	Total Revenue (Rs)	Markup from Carbon trading, consulting (15%)	Variable Costs (Rs)	Contribution Margin (Rs)
1 (60%)	3.8 lac	2 lac	1.8 lac	230 Cr	0	120 Cr	110 Cr
2 (80%)	3.8 lac	2 lac	1.8 lac	300 Cr	45 Cr	160 Cr	185 Cr
3 (90%)	3.8 lac	2 lac	1.8 lac	340 Cr	50 Cr	180 Cr	210 Cr



- Investment fixed: Rs 300 Cr for 10,000 tonne plant and to be expanded in future
- Achieve ~60% operating capacity in 1st yr and 80% and 90% thereafter in 2nd and 3rd yr
- Contribution margin is around 50%



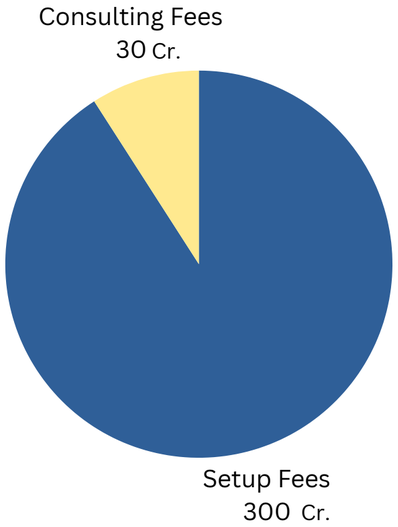
Breakeven Analysis



Opportunity for EY

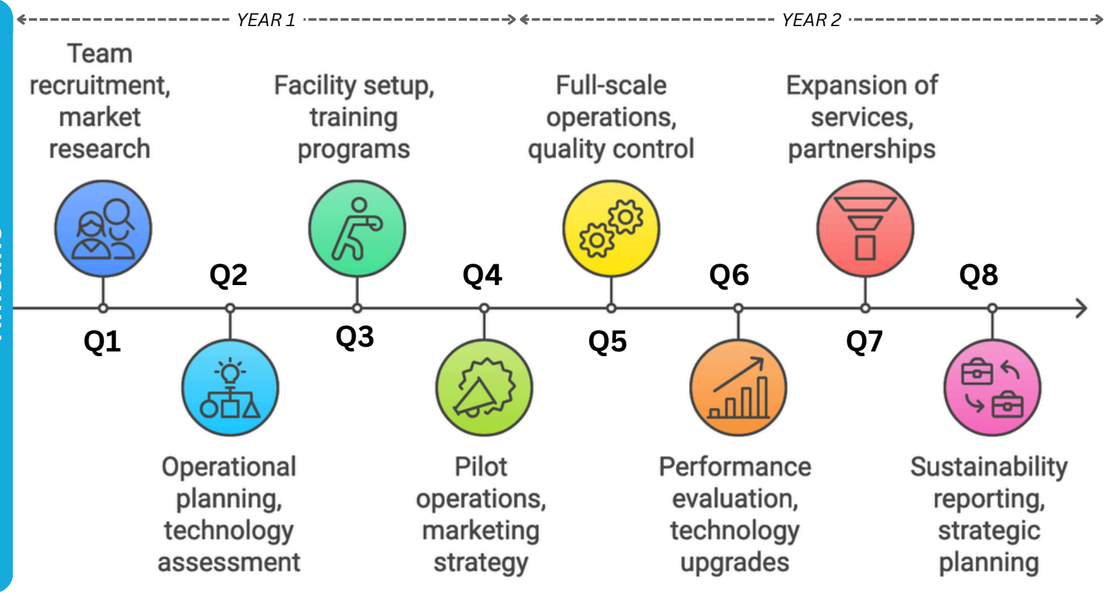
- Help setup the entire plan for firm and can charge upto 10% of setup fees as consulting fees so ~30 crores
- Continued support services for future expansion, driving economies of scale and achieving synergies
- **Category creation** so get headstart in industry and publish indepth industry reports and help government set policy

30 Cr.



(McKinsey) | (WRI India)

Timeline



People
- Team Structure
- Training
- Stakeholder Engagement

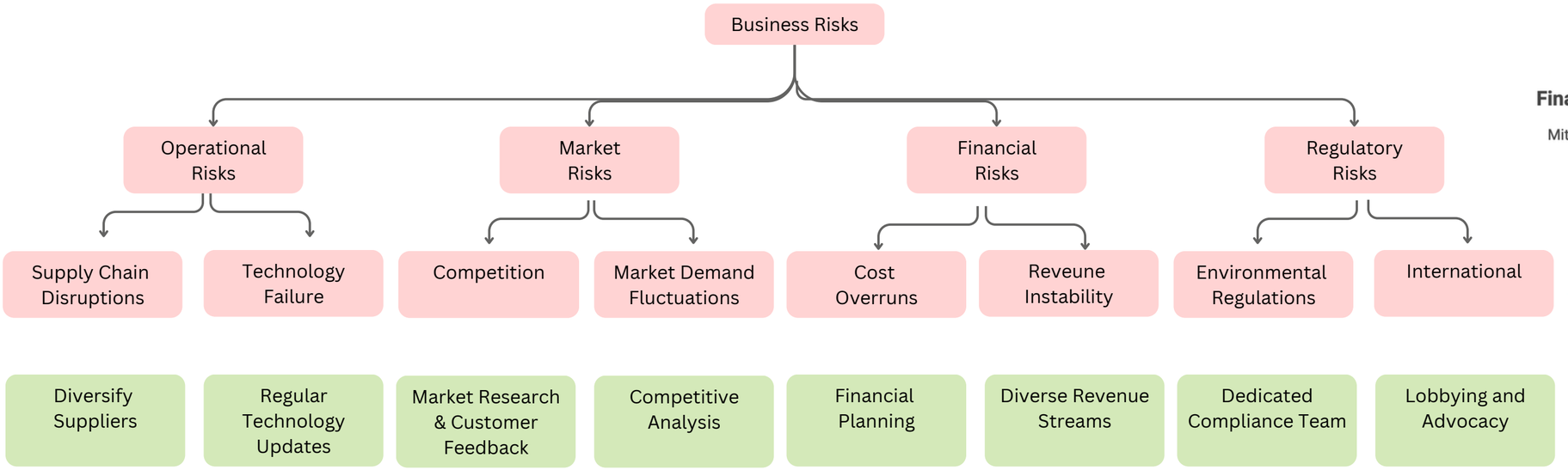
Process
- Workflow
  - Collection, sorting, recycling
- Compliance
- Sustainability

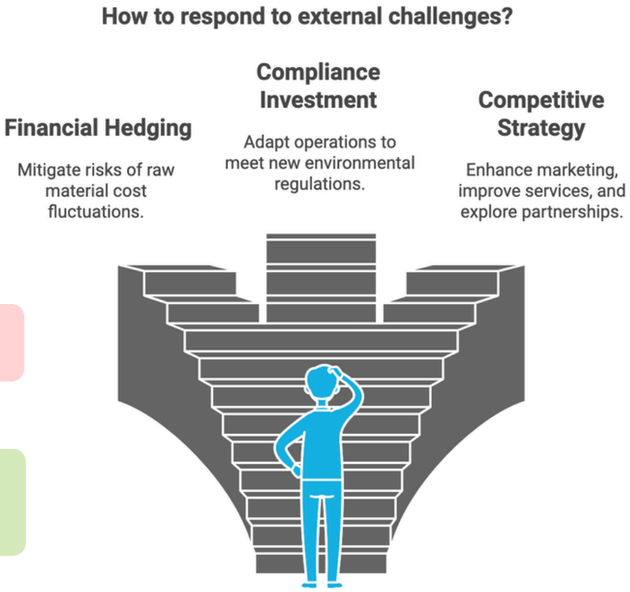
Technology
- Recycling Tech
- Data Management
- CRM
- R&D

The idea is to ensures a solid foundation and sustainable growth:
- **Year 1: Build infrastructure**—recruit team, secure technology, establish workflows, pilot operations.
- **Year 2: Scale up**—launch full operations with quality controls, expand services, strengthen partnerships, and report on sustainability.

This approach tries to minimise risk, maximises efficiency, and build credibility for long-term success.

Risk Identification & Mitigation





Rules and Regulations for Battery Recyclers

Registration and Reporting

- Registration: Apply for registration with the Ministry of Environment, Forest, and Climate Change.
- Annual Returns: Submit reports to the State Pollution Control Board by June 30th and December 31st.
- Record Keeping: Maintain records of battery receipts, sources, quantities, and metal yield, and submit them to the State Pollution Control Board.

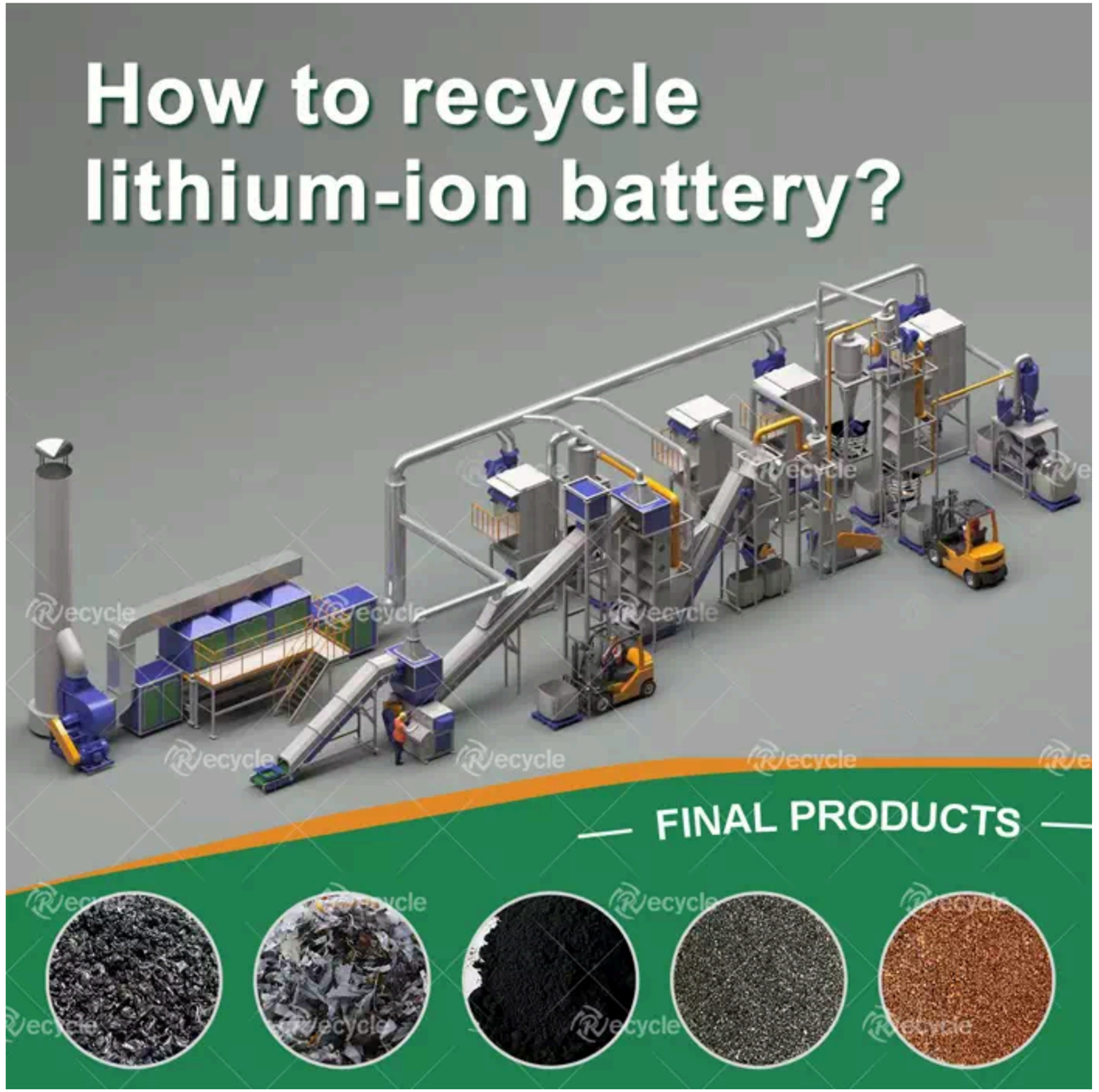
Recycling Processes and Labeling

- Metal Labeling: Mark recovered metals with "recycled."
- Capacity: Central recycling units should have a capacity of over 10,000 tons/year.
- Adherence to Terms: Ensure strict compliance with registration terms and conditions.

Public Awareness

- Awareness Campaigns: Create awareness about battery recycling through advertisements, publications, and other means.
- Responsible Disposal: Promote the importance of returning used batteries to authorized recyclers.
- Hazard Education: Explain the dangers of heavy metals in batteries.

Government Scheme/Initiative	Description	Impact on Battery Recycling Business
FAME India Scheme (Phase II)	Provides subsidies to electric vehicle buyers, promoting the adoption of EVs.	Increases the volume of batteries for recycling as EV usage expands.
Battery Waste Management Rules, 2022	Mandates proper disposal, recycling, and refurbishment of batteries under the Extended Producer Responsibility (EPR).	Directly boosts the recycling sector by ensuring a supply of waste batteries.
Production Linked Incentive (PLI) for ACC	Incentivizes domestic manufacturing of Advanced Chemistry Cells (ACCs) to boost local industry.	Encourages the establishment of recycling facilities to support ACC production.
Resource Efficiency and Circular Economy Industry Coalition (RECEIC)	Aims to enhance resource efficiency and promote a circular economy across various sectors.	Supports recycling initiatives and integrates them into broader economic policies.
Swachh Bharat Mission	Focuses on cleanliness and waste management across India.	Encourages proper disposal and recycling of batteries as part of waste management.
Atmanirbhar Bharat Abhiyan	A self-reliance initiative that includes boosting domestic industries.	Promotes local recycling technologies and reduces dependence on battery imports.
National Electric Mobility Mission Plan	Aims to achieve national fuel security by promoting hybrid and electric vehicles.	Increases demand for battery recycling services as more EVs enter the market.


How to recycle
lithium-ion battery?
FINAL PRODUCTS

The Economic Times

India set to open first lithium mines in Chhattisgarh

India will open its first lithium mines in the Katghora region of Korba district, Chhattisgarh, following the Geological Survey of India's...

14 Aug 2024

India Today

Lithium deposits estimated at 1,600 tonnes found in Karnataka

The Atomic Minerals Directorate for Exploration and Research (AMD), a unit of the Department of Atomic Energy, has identified 1600 tonnes of...

26 Jul 2024

ETEnergyworld.com

India's battery energy storage capacity grows over 4-fold to 219 MW in Q1: Mercom Capital

New Delhi: India's battery energy storage capacity grew more than four-fold to 219 MWh as of March 2024, Mercom Capital said.

10 Jul 2024

mint

Govt scientific advisor wants lithium alternatives explored amid global race | Mint

Sood specifically mentioned sodium-ion and aluminium air batteries as alternatives to lithium.

4 Jan 2024

Autocar Professional

India to become the largest lithium-ion battery exporter in 5 years: Nitin Gadkari

India has the potential to become the largest lithium-ion battery exporter globally by the end of the decade, according to the Minister of Road Transport and...

5 Sept 2024

Business Standard

Govt considers offering incentives for lithium processing: Report

India is considering offering incentives to encourage private companies to set up lithium processing facilities, as New Delhi tries to...

9 May 2024

Business Standard

Govt introduces strict EC guidelines for violations of battery waste rules

The Ministry of Environment, Forest, and Climate Change (MoEFCC) has introduced stringent environmental compensation (EC) guidelines to penalise violations of...

16 Sept 2024

THANK YOU